Protamine sulfate: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Protamine sulfate: Pediatric drug information" and "Protamine sulfate: Patient drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Hypersensitivity reactions:

Protamine sulfate can cause severe hypotension, cardiovascular collapse, noncardiogenic pulmonary edema, catastrophic pulmonary vasoconstriction, and pulmonary hypertension. Risk factors include high dose or overdose, rapid administration, repeated doses, previous administration of protamine, and current or previous use of protamine-containing drugs (NPH insulin, protamine zinc insulin, and certain beta-blockers). Allergy to fish, previous vasectomy, and severe left ventricular dysfunction and abnormal preoperative pulmonary hemodynamics also may be risk factors. In patients with any of these risk factors, the risk to benefit of administration of protamine sulfate should be carefully considered. Vasopressors and resuscitation equipment should be immediately available in case of a severe reaction to protamine. Protamine sulfate should not be given when bleeding occurs without prior heparin use.

Pharmacologic Category

- Antidote

Dosing: Adult

The adult dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editor: Edith A Nutescu, PharmD, MS, FCCP.

Heparin neutralization

Heparin neutralization:

Following *IV* heparin administration: IV: Initial: Protamine dosage is determined by the amount of heparin administered; 1 mg of protamine neutralizes ~100 units of heparin; maximum single dose: 50 mg. Administer by slow IV injection not to exceed 5 mg/minute (Ref); if additional doses are needed (eg, aPTT or anti–factor Xa remains elevated), may administer 0.5 mg of protamine for every 100 units of heparin (Ref).

Note: Because heparin concentration decreases rapidly after administration (half-life of heparin is ~60 to 90 minutes), adjust protamine dosage depending on duration of time since heparin administration. For example, if 2 hours has elapsed since a heparin overdose, administer half of the calculated initial protamine dose (Ref). If heparin was administered as a continuous IV infusion, calculate protamine dose based on heparin administered in the preceding 2 to 3 hours. For example, a patient receiving heparin 1,250 units/hour will require ~30 mg of protamine to neutralize heparin (Ref). If patient is **not** bleeding, consider **not** administering protamine since risks may outweigh benefits (Ref).

Cardiac surgery patients: May consider lower doses (eg, 0.5 mg for 100 units of heparin) to prevent excess protamine administration (Ref). After cardiopulmonary bypass (CPB), repeat protamine doses of 25 to 100 mg may be given to reverse large doses of intraoperative heparin if activated clotting time (ACT) remains elevated or if heparin rebound is a concern; maximum total dose: 3 mg/kg (Ref); however, ACT values may not consistently correlate with unfractionated heparin levels during or after CPB. For heparin rebound, may consider protamine 25 mg/hour continuous IV infusion for 4 to 6 hours following the initial dose (Ref).

Following *SUBQ* heparin injection: Note: May consider protamine to neutralize prophylactic SUBQ doses of heparin when aPTT is significantly prolonged and patient has clinically significant bleeding (Ref).

IV: Initial: Protamine dosage is determined by the amount of heparin administered; 1 mg of protamine neutralizes ~100 units of heparin; administer by slow IV injection over ~10 minutes; maximum single dose: 50 mg.

Note: Consider heparin absorption via SUBQ route when determining protamine dose. A portion of the protamine dose may be given IV over 10 minutes followed by the remaining portion as a continuous infusion over 8 to 16 hours (the expected absorption time of the SUBQ heparin dose) (Ref). If patient is **not** bleeding, consider **not** administering protamine since risks may outweigh benefits (Ref).

Low-molecular-weight heparin neutralization

Low-molecular-weight heparin neutralization (off-label use): Note: Protamine will not completely neutralize anti–factor Xa activity (maximum: ~60% to 75%). Excessive protamine doses may worsen bleeding (Lovenox prescribing information). Consider using in patients with clinically significant bleeding. If the patient is **not** bleeding, consider **not** administering protamine since risks may outweigh benefits (Ref).

IV:

Enoxaparin:

Enoxaparin administered in ≤8 hours: Administer 1 mg of protamine for every 1 mg of enoxaparin administered. May repeat with an additional dose of 0.5 mg of protamine for every 1 mg of enoxaparin, as needed (Ref). Administer by slow IV injection (eg, 50 mg over ~10 minutes); maximum: 50 mg per dose (Ref).

Enoxaparin administered >8 hours to <12 hours ago: Administer 0.5 mg of protamine for every 1 mg of enoxaparin administered. May repeat with an additional dose of 0.5 mg of protamine for every 1 mg of enoxaparin, as needed (Ref). Administer by slow IV injection (eg, 50 mg over ~10 minutes); maximum: 50 mg per dose (Ref).

Enoxaparin administered ≥12 hours ago (Ref) *or if 3 to 5 half-lives have elapsed* (Ref)*:* Protamine administration may not be required.

Dalteparin, nadroparin, or tinzaparin: 1 mg of protamine for every 100 anti–factor Xa units of low-molecular-weight heparin (LMWH) administered within the past 3 to 5 half-lives; administer by slow IV injection over ~10 minutes; maximum single dose: 50 mg. If clinically significant bleeding persists or patient has renal impairment, consider repeat dose of 0.5 mg of protamine for every 100 anti–factor Xa units of LMWH (Ref).

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Protamine sulfate: Pediatric drug information")

Dosage guidance:

Clinical considerations: 1 mg of protamine sulfate neutralizes ~100 units of heparin or 1 mg of enoxaparin (Ref).

Heparin neutralization/reversal

Heparin (intravenous) neutralization/reversal: Limited data available:

Infants, Children, and Adolescents: IV: **Note:** Since heparin disappears rapidly from the circulation, the dose of protamine is adjusted based upon the time since heparin administration (see table). If given as a continuous IV infusion, only heparin given in the preceding 2 to 4 hours should be considered when administering protamine (Ref).

Table 1: **Heparin Neutralization/Reversal**

Time Since Last Heparin Dose	Dose of Protamine
<30 minutes	IV: 1 mg of protamine per 100 units of heparin; maximum dose: 50 mg/dose
30 to 60 minutes	IV: 0.5 to 0.75 mg of protamine per 100 units of heparin; maximum dose: 50 m
60 to 120 minutes	IV: 0.375 to 0.5 mg of protamine per 100 units of heparin; maximum dose: 50
>120 minutes	IV: 0.25 to 0.375 mg of protamine per 100 units of heparin; maximum dose: 50

Low molecular weight heparin neutralization/reversal

Low molecular weight heparin (LMWH) neutralization/reversal (enoxaparin, dalteparin):

Note: Protamine will not completely neutralize the anti-Xa activity (maximum: ~60% to 75%). Protamine dosage is determined by the most recent dosage of LMWH (Ref):

Enoxaparin (intravenous and subcutaneous) neutralization/reversal: Limited data available: Dosing extrapolated from experience in adult patients:

Infants, Children, and Adolescents: IV: Protamine dosage is determined by the most recent dosage and time of administration of enoxaparin (Ref).

Enoxaparin dose administered ≤8 hours: IV: Initial: 1 mg of protamine for every 1 mg of enoxaparin administered; if needed, may give a second dose of 0.5 mg of protamine for every 1 mg of enoxaparin administered (ie, prolonged aPTT measured 2 to 4 hours after initial dose) (Ref).

Enoxaparin administered >8 hours prior or if it has been determined that a second dose of protamine is required (eg, if aPTT measured 2 to 4 hours after the first dose remains prolonged or if bleeding continues): IV: Initial: 0.5 mg protamine sulfate for every 1 mg enoxaparin administered; if needed, may repeat with a second dose of 0.5 mg of protamine for every 1 mg of enoxaparin administered (ie, prolonged aPTT measured 2 to 4 hours after initial dose) (Ref).

Dalteparin (subcutaneous) neutralization/reversal:

Infants, Children, and Adolescents: IV: 1 mg protamine for each 100 anti-Xa units of dalteparin; if aPTT prolonged 2 to 4 hours after first dose (or if bleeding continues), consider additional dose of 0.5 mg for each 100 anti-Xa units of dalteparin (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Flushing

Gastrointestinal: Nausea, vomiting

Local: Localized warm feeling

Nervous system: Lassitude

Respiratory: Dyspnea

Postmarketing:

Cardiovascular: Bradycardia (McDonald 2020), circulatory shock, hypotension (McDonald 2020), right heart failure (right ventricular failure) (Pannu 2016), severe hypotension (Leung 2019, Pannu 2016), ventricular fibrillation (Leung 2019)

Hematologic & oncologic: Immune thrombocytopenia (Singla 2013)

Hypersensitivity: Hypersensitivity reaction (including anaphylactic shock, anaphylaxis, angioedema) (Doolan 1981, Roelofse 1991), nonimmune anaphylaxis

Neuromuscular & skeletal: Back pain

Respiratory: Noncardiogenic pulmonary edema (Kindler 1996), pulmonary complications (pulmonary vasoconstriction) (Lowenstein 1983), pulmonary hypertension (can be acute) (Pannu 2016)

Contraindications

Hypersensitivity to protamine or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

• Heparin rebound: Heparin rebound associated with anticoagulation and bleeding has been reported to occur occasionally; symptoms typically occur 8-9 hours after protamine administration, but may occur as long as 18 hours later.

• Hypersensitivity reactions: May cause hypersensitivity reaction in patients (have appropriate treatment available).

• Infusion reactions: Too rapid administration can cause severe hypotensive and anaphylactoid-like reactions.

Special populations:

• Cardiac surgery patients: May be ineffective in some patients following cardiac surgery despite adequate doses.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as sulfate:

Generic: 10 mg/mL (5 mL, 25 mL)

Solution, Intravenous, as sulfate [preservative free]:

Generic: 10 mg/mL (5 mL, 25 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Protamine Sulfate Intravenous)

10 mg/mL (per mL): $2.68 - $4.23

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as sulfate:

Generic: 10 mg/mL (5 mL, 25 mL)

Administration: Adult

IV: For IV use only. Administer slow IVP (eg, over 10 minutes). Rapid IV infusion causes hypotension; maximum of 50 mg in any 10-minute period.

Preparation for Administration: Adult

May be further diluted in D5W or NS.

Administration: Pediatric

Parenteral: IV: Administer undiluted (preferred) by slow IVP at a rate not to exceed 5 mg/minute (maximum rate: 50 mg in any 10-minute period); may be administered as diluted solution. **Note:** Rapid IV infusion causes hypotension.

Preparation for Administration: Pediatric

May be further diluted in D5W or NS.

Storage/Stability

Store at 20°C to 25°C (68°F to 77°F). Do not freeze. Diluted solutions should not be stored.

Use: Labeled Indications

Heparin neutralization: Treatment of heparin overdosage.

Use: Off-Label: Adult

Low-molecular-weight heparin neutralization

Medication Safety Issues

Sound-alike/look-alike issues:

Protamine may be confused with ProAmatine, Protonix, Protopam

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

There are no known significant interactions.

Pregnancy Considerations

Animal reproduction studies have not been conducted. In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant women if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey, 2003). Protamine sulfate may be used during delivery to reduce the risk of bleeding following maternal use of heparin or low molecular weight heparin (LMWH) (Bates, 2012).

Breastfeeding Considerations

It is not known if protamine is excreted in breast milk. The manufacturer recommends that caution be exercised when administering protamine to nursing women.

Monitoring Parameters

Coagulation tests, hemodynamics during administration.

Mechanism of Action

Protamine, a highly alkaline protein molecule with a large positive charge, has weak anticoagulant activity when administered alone. When protamine is given in the presence of heparin (strongly acidic and negatively charged), a stable salt is formed and the anticoagulant activity of both drugs is nullified (Pai 2012). In the presence of LMWH, protamine incompletely reverses the anti–factor Xa activity of LMWH (Makris 2000; Massonnet-Castel 1986; Racanelli 1985).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV: Heparin neutralization: ~5 minutes

Half-life elimination: ~7 minutes

Patient Counseling Points

(For additional information see ”Protamine sulfate: Patient drug information”)

What is this drug used for?

• It is used to treat heparin overdose.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Loss of strength and energy

• Flushing

• Sensation of warmth

• Back pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Shortness of breath

• Slow heartbeat

• Severe dizziness

• Passing out

• Nausea

• Vomiting

• Bruising

• Bleeding

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Prosulf | Protamine sulphate;
- (AR) Argentina: Denpru | Protamina;
- (AU) Australia: Protamine sulphate;
- (BE) Belgium: Protamine | Protamine sulfaat leo;
- (BG) Bulgaria: Protamin sulfat leo;
- (CL) Chile: Protamina sulfato;
- (CR) Costa Rica: Protamina sulfato;
- (CZ) Czech Republic: Protamin | Protamin sulfate;
- (DO) Dominican Republic: Protamina;
- (EC) Ecuador: Protamina | Protamina sulfato;
- (EE) Estonia: Prosulf | Protamin | Protamin sulphate;
- (EG) Egypt: Protamine sulphate | Sulfiprotam;
- (ES) Spain: Protamina leo | Protamina rovi | Protamina sulfato leo pharma;
- (FI) Finland: Protamiinisulfaatti Leo Pharma | Protamin | Protamini sulfas | Protaminsulfat Leo pharma;
- (FR) France: Prosulf | Protamine choay;
- (GB) United Kingdom: Prosulf | Protamin sulphate | Protamine sulphate;
- (HK) Hong Kong: Prosulf;
- (ID) Indonesia: Protamine sulphate;
- (IE) Ireland: Protamine sulphate;
- (IN) India: Neutrahep | Orvoprot;
- (IT) Italy: Protamina | Protamina solfato | Protamina solfato leo pharma;
- (JP) Japan: Novo protamin sulfate | Novo protamine sulfate | Protami.sulf.ajino;

- (KR) Korea, Republic of: Urota;
- (KW) Kuwait: Prosulf;
- (LB) Lebanon: Prosulf | Protamine;
- (LT) Lithuania: Protamin sulfate | Protamine sulphate;
- (LU) Luxembourg: Protamine Sulfaat;
- (LV) Latvia: Protamin sulphate | Protamine sulphate;
- (MA) Morocco: Protamine choay;
- (MY) Malaysia: Prosulf | Protamine sulphate | Taminfate;
- (NL) Netherlands: Protamine sulfaat LEO Pharma;
- (NZ) New Zealand: Protamine sulphate;
- (PA) Panama: Protamina sulfato;
- (PE) Peru: Protamina sulfato | Protaxat;
- (PL) Poland: Neutrahep | Prosulf | Protamin sulfat | Protamine | Protamine sulphate;
- (PT) Portugal: Protamina;
- (QA) Qatar: Promin | Prosulf | Protamin ICN;
- (RO) Romania: Pamintu | Protaminsulfat Leo pharma;
- (RU) Russian Federation: Protagem | Protamin | Protamin Ferein | Protamine;
- (SE) Sweden: Protamin sulfat | Protaminsulfat Leo pharma | Protaminsulfat paranova;
- (SG) Singapore: Protamine sulphate;
- (SI) Slovenia: Protamin sulfat | Protaminijev sulfat;
- (SK) Slovakia: Protamin sulfate;
- (TN) Tunisia: Prosulf;
- (TR) Turkey: Pamintu | Promin;
- (TW) Taiwan: Protamine sulphate;
- (UA) Ukraine: Protamine sulphate;
- (UY) Uruguay: Denpru | Protamina;
- (ZA) South Africa: Protamine sulphate

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